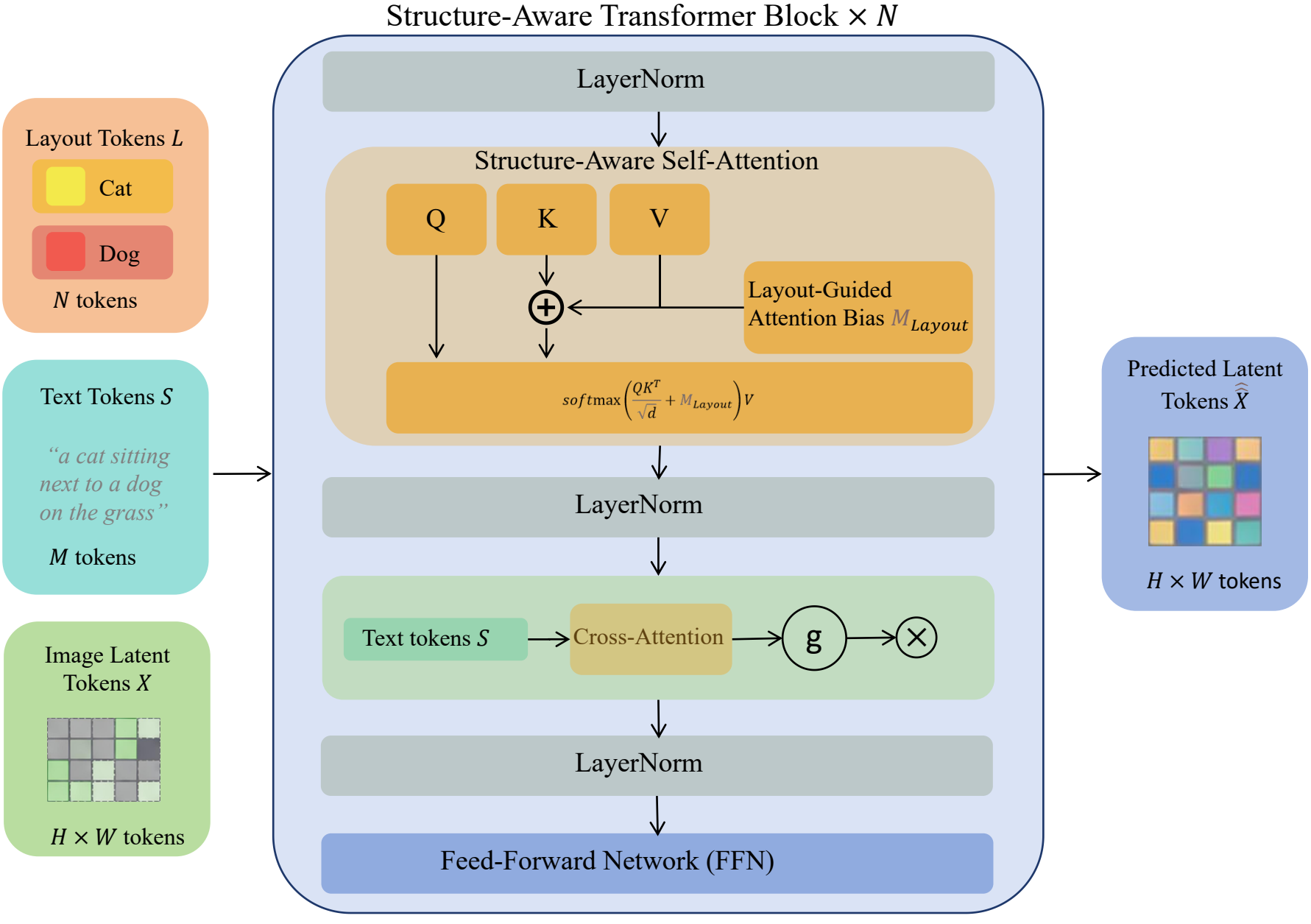
Structure-Aware Transformer Block × N
Layout Tokens L
Cat
Dog
N tokens
Text Tokens S
"a cat sitting next to a dog on the grass"
M tokens
Image Latent Tokens X
H × W tokens
LayerNorm
Structure-Aware Self-Attention
Q
K
V
Layout-Guided Attention Bias M_{Layout}
$softmax\left(\frac{QK^T}{\sqrt{d}} + M_{Layout}\right)V$
LayerNorm
Text tokens S
Cross-Attention
g
LayerNorm
Feed-Forward Network (FFN)
Predicted Latent Tokens $\hat{\hat{X}}$
H × W tokens